

1. (Fall 2024, #9) Evaluate the following integrals.

(a) $\int_1^e \frac{\ln x}{x\sqrt{1+(\ln x)^2}} dx$

(b) $\int e^{xe^x}(e^x + xe^x)dx$

2. (Spring 2024, #8) Evaluate the following integrals.

(a)

$$\int \tan(x) \ln(\cos(x)) dx$$

(Hint: what is the simplified derivative of $\ln(\cos(x))$?)

(b)

$$\int_0^1 \frac{2x + e^{3x}}{3x^2 + e^{3x} + 3} dx$$

3. (Fall 2023, #8) Compute the following indefinite/definite integrals.

(a) $\int \frac{x^{3/2} - 5x + 2}{x^{1/2}} dx$

(b) $\int \frac{x}{\sqrt{x-2}} dx$

(c) $\int_0^{\ln 5} \frac{e^x dx}{2e^x + 5}$

4. (Spring 2023, #8) Evaluate the following integrals.

(a) $\int \frac{\cot(x)}{\ln(\sin(x))} dx$

(b) $\int_0^{\ln(\pi)} \left(e^{3e^{3x}+3x} \right) dx$

5. (Fall 2022, #7) Find the following explicitly:

(a) $\int_0^4 |x - 2| dx$

(b) $\int_0^{1/2} x \sec^2(\pi x^2) dx$

(c) $\int \frac{1}{x \ln(x)} dx$